MANIPAL UNIVERSITY

INSPIRED BY LIFE

School of Computer Science and Engineering

Department of Computer Science and Engineering

UNI LOST

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Outline

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Introduction

Frequent Item Loss

College students frequently lose important items such as ID cards, wallets, essential books, and expensive gadgets, leading to immediate stress and disruption.

Manual & Unorganized Processes

Current manual solutions (e.g., paper registers, bulletin boards, informal chat groups) are inherently slow, unorganized, and prone to misplacement of reports.

Consequences of Delay

Delayed recovery results in significant inconvenience, potential loss of important documents, academic setbacks, and widespread user frustration across the campus community.

Introducing the Digital Lost & Found System

We propose a centralized, web-based platform designed to drastically minimize the time and effort required to locate lost belongings and ensure trust in the recovery process.

Web-Based Platform

Accessible via any browser, simplifying reporting and searching for all students and administrators.

Image Upload & Match

Users can upload images for visual identification and view automated suggestions for potential matches instantly.

Verification & Security

Admin verification ensures the authenticity of all claims, ensuring items are returned to their rightful owners.

Literature Review

A comparative analysis of existing methods highlights the clear need for a digitally centralized and automated solution for the campus environment.

Platform	Approach	Limitation
Manual Registers	Paper-based entry	Time-consuming, non-searchable, and easily lost or damaged.
WhatsApp Groups	Informal, chat-based reporting	Lack of formal tracking, verification, and data retention.
Google Forms	Basic form input for data collection	No automated matching algorithm or administrative control over claims.

Observation: Current systems are either inefficient, lack centralized tracking, or require excessive manual effort to ma claims.

The Gaps in Current Lost & Found Processes

- **No Automated Matching**

Lost item reports are not automatically compared against found item reports, relying instead on manual scanning.

- **Lack of Image Verification**

Identification relies solely on subjective textual descriptions, leading to potential disputes or false claims.

- **Decentralized Records**

Information is scattered across various departments and platforms, making comprehensive searching impossible.

- **Missing Administrative Control**

Absence of a formal process for ownership verification and claim approval.

Problem Statement

"Students face difficulty locating lost belongings due to the absence of a verified and centralized digital Lost & Found system within the campus."

The core problem lies in the inefficiency of current systems, where manual or unstructured methods lead to lost records, opportunities for false claims, and substantial user inconvenience. We need a structured platform that guarantees authenticity and efficiency for all stakeholders.

Objectives

→ **User-Friendly Web Portal**

Develop an intuitive portal for seamless reporting and management of lost and found items by students and faculty.

→ **Scalable NoSQL (MongoDB) Database**

Store all item and user data securely in a robust MongoDB Atlas cloud database, ensuring flexible data retrieval, high availability, and long-term record keeping.

→ **Automated Matching Logic**

Implement logic to automate search and matching based on keywords, category, and time, reducing manual effo and errors.

→ **Authenticity Verification**

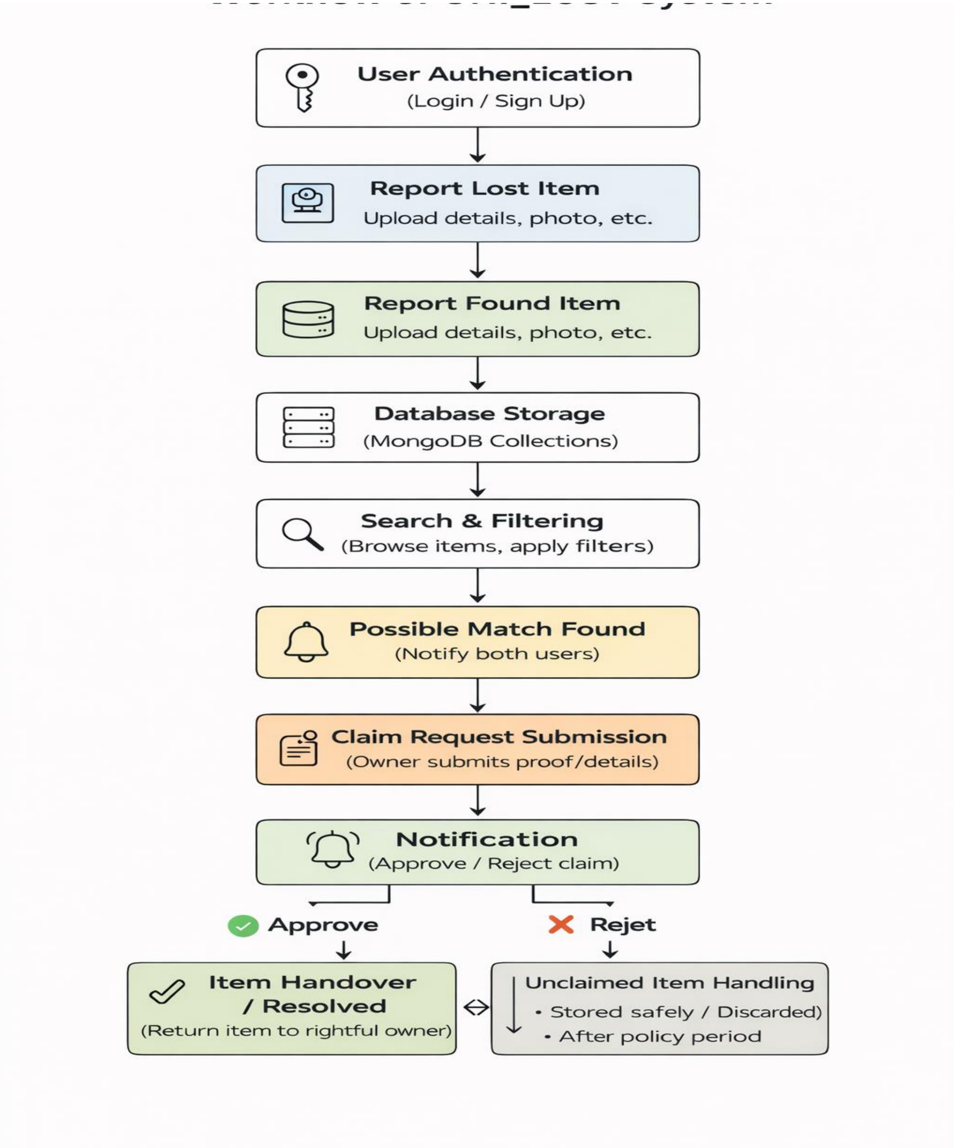
Establish an admin verification process to thoroughly vet all claims and ensure items are only returned to the rightful owners.

→ **Detailed Reporting & Statistics**

Provide administrators with detailed reports to track trends in lost/found items and measure system effectiveness

Proposed Methodology

- FLOW CHART



Result

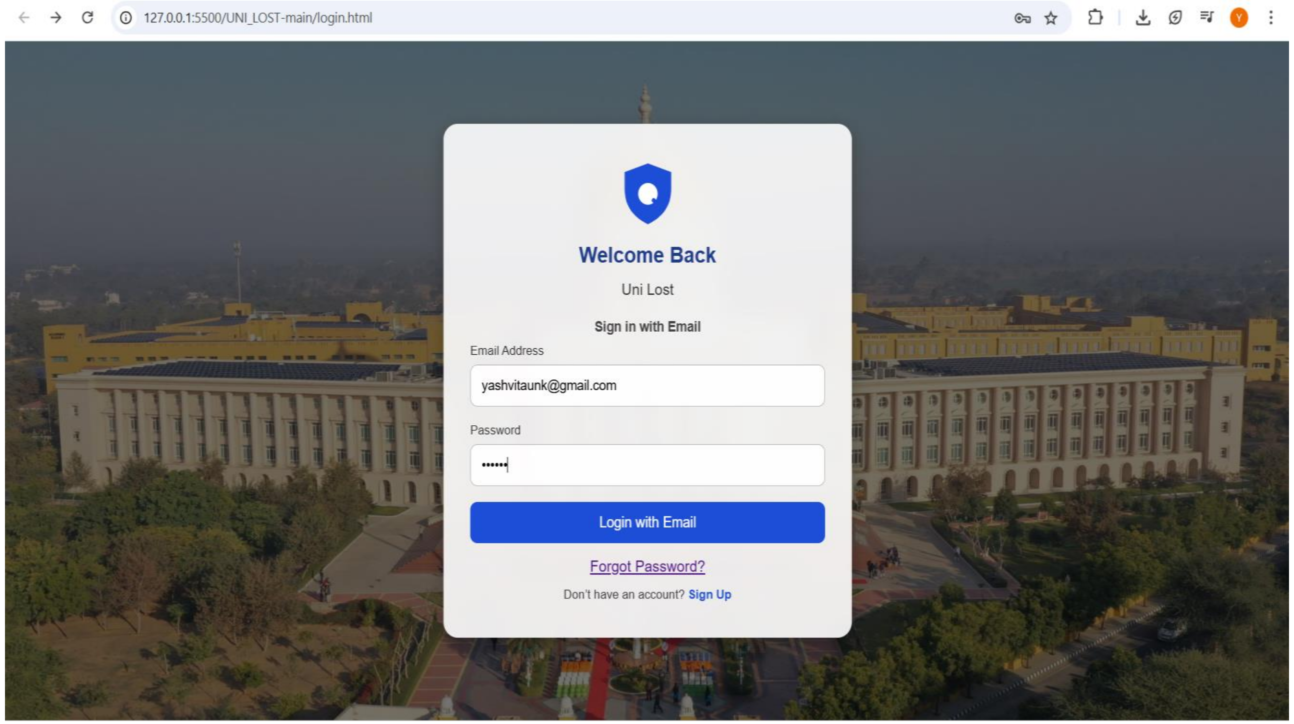
Website Frontend (UI/UX)

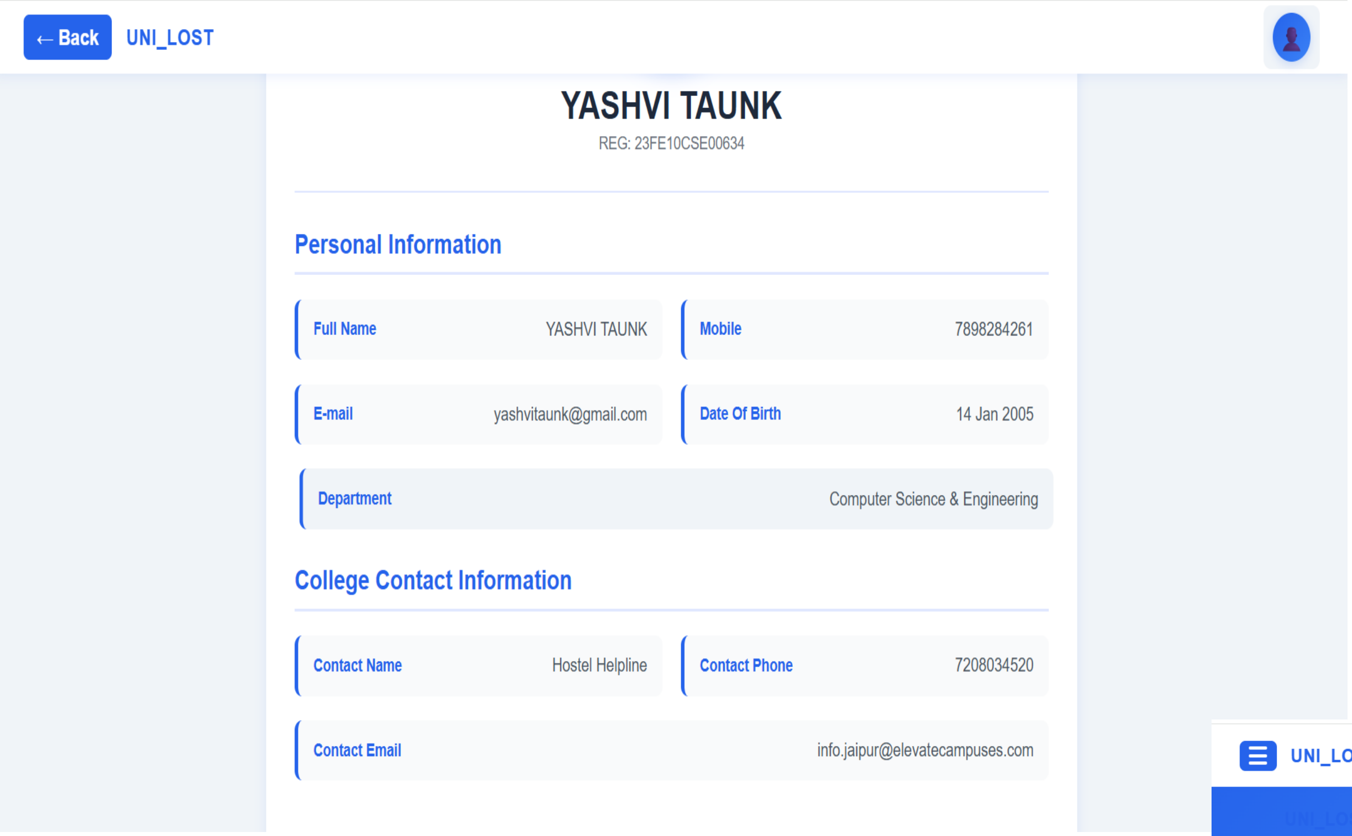
1. **Seamless User Experience:** Developed a responsive and intuitive interface for easy navigation and item reporting.

2. **Real-time Interaction:** Users can register, log in, and post lost or found items instantly.

1. **Dynamic Search & Filter:** Integrated functionality to filter items by category and location for faster recovery.

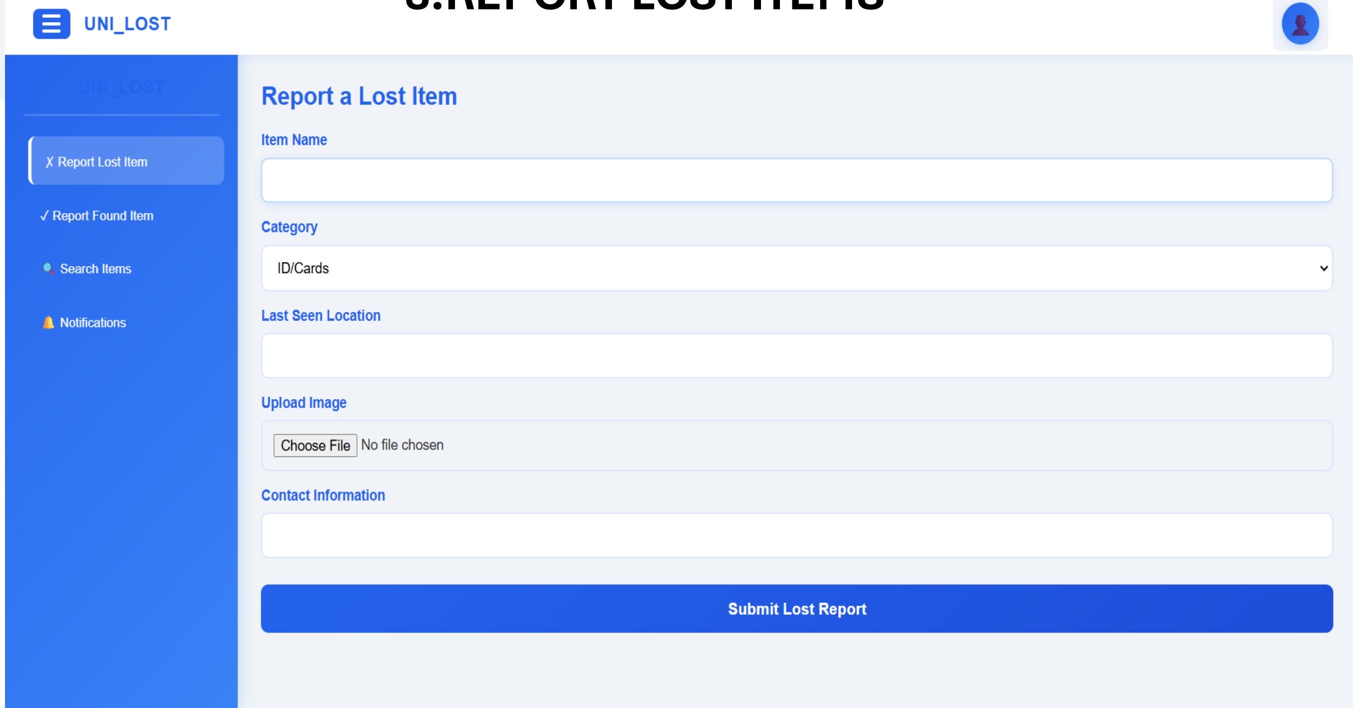
1..SIGNUP/ LOGIN PAGE (USER AUTEHNTICATION)

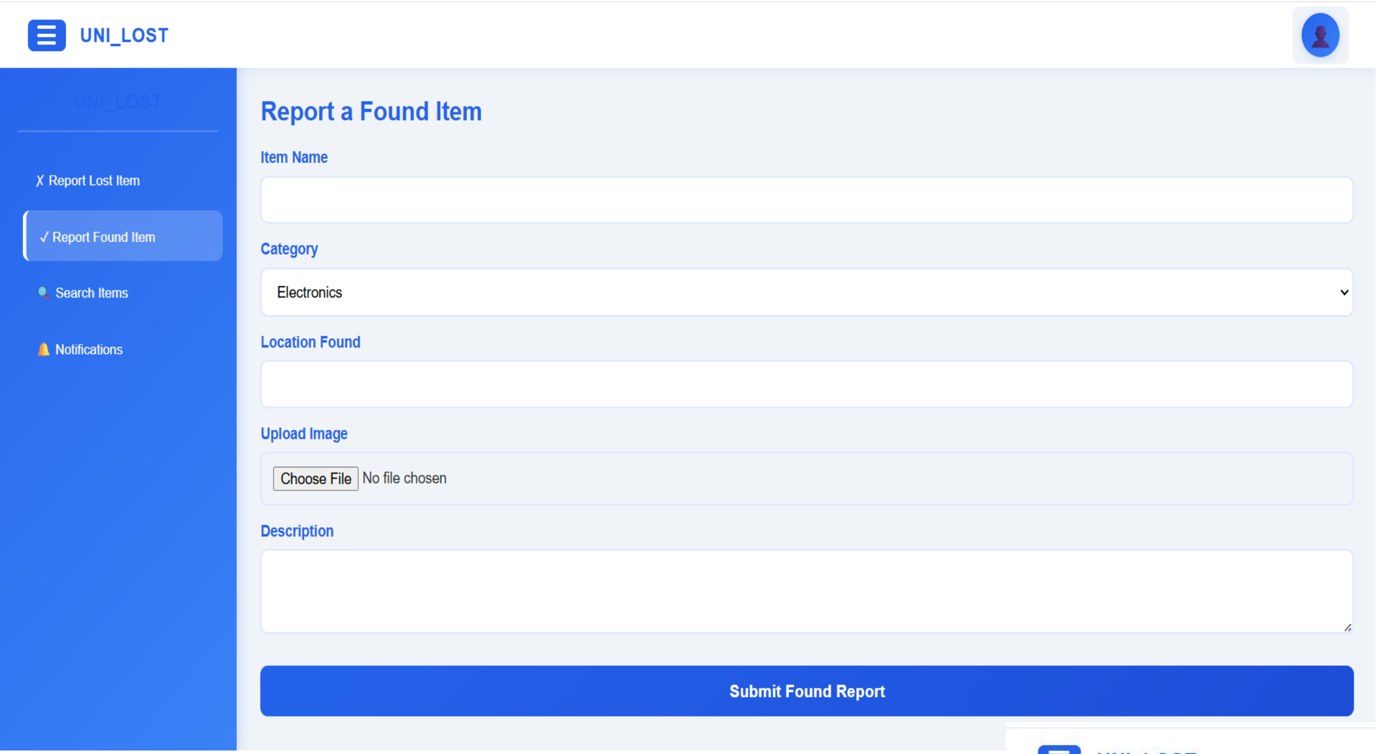




2.PROFILE SECTION (USER DETAILS)

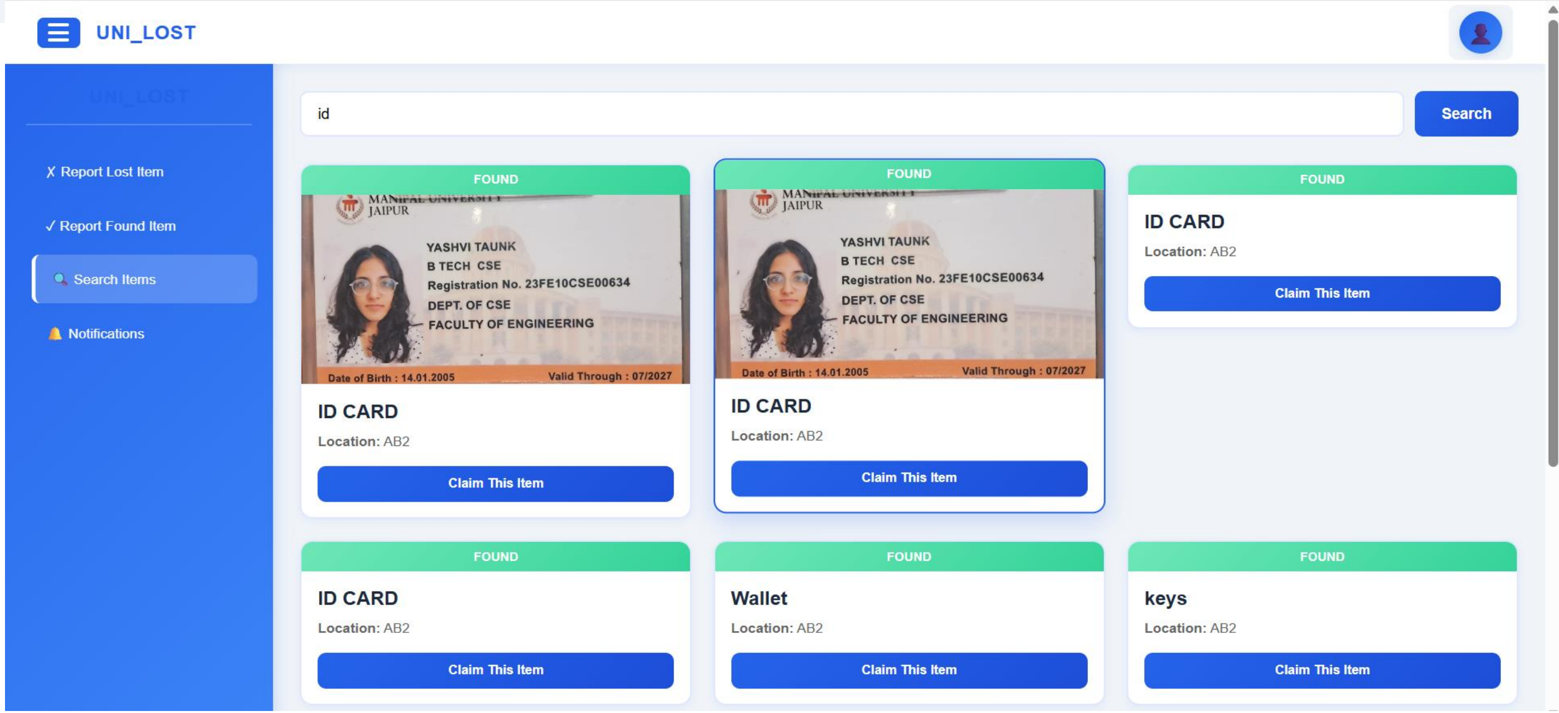
3.REPORT LOST ITEMS

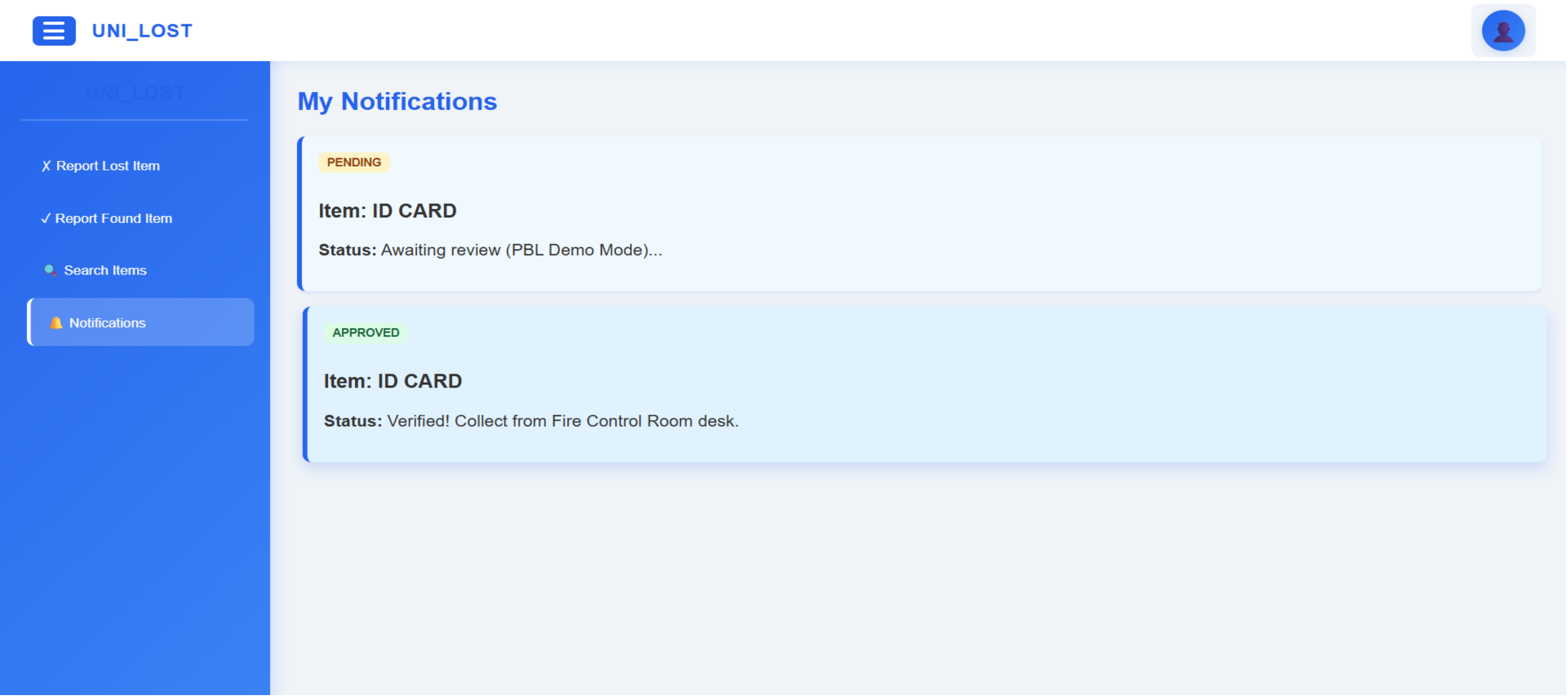




4.REPORT LOST ITEMS

5. SEARCH ITEMS





6. NOTIFICATION

Backend & Database (Security & Storage)

- **Cloud Database Integration:** Uses MongoDB Atlas for secure, scalable cloud storage of users, items, and reports.
- **Secure Authentication:** Passwords are encrypted using bcryptjs hashing to ensure user data protection.
- **RESTful API Architecture:** Efficient API communication between frontend and backend for smooth data handling.
- **Deployment on Render:** Backend services are deployed on Render for reliable hosting, continuous deployment, and real-time accessibility.

Outcome

- **Successful Digitalization :** Successfully replaced inefficient, paper-based processes with a modern, fast, and user-friendly web application.
- **Enhanced Student Experience:** Reduced the stress and time required for students to report or search for lost property, improving overall campus morale.
- **Seamless Information Flow:** Implemented a single-page Dashboard interface ensuring rapid navigation between key actions: Searching, Reporting Lost, and Reporting Found items.
- **Integrated Security:** Incorporated robust client-side validation for login and detailed personal profile integration (Reg. No., Contact, Course) via a Modal Pop-up to streamline verification during the claim process.
- **Future-Ready Platform:** The modular design (using HTML, CSS, and JS) allows for easy integration with future database/backend systems for scaling and permanent data storage.

References

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Thank You